

$$(14) (i) \lim_{x \rightarrow 0} \frac{\tan x}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \times \lim_{x \rightarrow 0} \frac{1}{\cos x}$$

$$1 \times \frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \cos x}$$

$$1 \times 1$$

$$1 //$$

$$(15) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{\frac{x^2}{2}}$$

$$24 \times \frac{x^2}{4}$$

$$\frac{1}{2} \left[\lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \right]^2$$

$$\frac{1}{2} \times 1 = \frac{1}{2} //$$

$$(ii) \lim_{x \rightarrow \infty} x \left(\sin \frac{1}{x} \right)$$

$$\lim_{x \rightarrow \infty} \frac{\sin \left(\frac{1}{x} \right)}{\frac{1}{x}}$$

$$\lim_{x \rightarrow \infty} \cos \left(\frac{1}{x} \right) \times \left(-\frac{1}{x^2} \right)$$

$$\left(-\frac{1}{x^2} \right)$$

$$\lim_{x \rightarrow \infty} \cos \frac{1}{x} = 1 //$$

$$(iv) \lim_{x \rightarrow 0} \frac{\tan x - x}{\sin x}$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{\sin x} - \lim_{x \rightarrow 0} \frac{x}{\sin x}$$

$$\lim_{x \rightarrow 0} \frac{1}{\cos x} - \frac{1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}}$$

$$1 - 1$$

$$0 //$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2\left(\frac{3x}{2}\right)}{\frac{9x^2}{4}} \times \frac{9}{4}$$

$$2 \times \frac{9}{4} \times \left[\lim_{x \rightarrow 0} \frac{\sin\left(\frac{3x}{2}\right)}{\frac{3x}{2}} \right]^2$$

$$\frac{9}{2} \times 1 = \frac{9}{2}$$

$$(vi) \lim_{x \rightarrow 0} \frac{\sin 4x + \tan 9x}{5x}$$

$$\lim_{x \rightarrow 0} \frac{\sin 4x}{5x} + \lim_{x \rightarrow 0} \frac{\tan 9x}{5x}$$

$$\frac{4}{5} \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} + \frac{9}{5} \lim_{x \rightarrow 0} \frac{\sin 9x}{9x}$$

$$= \frac{4}{5} \times 1 + \frac{9}{5} \times 1 \times 1$$

$$\frac{13}{5}$$

$$(vii) \lim_{x \rightarrow \pi/2} \frac{1 - \sin x}{\cot^2 x}$$

$$\lim_{x \rightarrow \pi/2} \frac{(1 - \sin x)}{\csc^2 x - 1}$$

$$\lim_{x \rightarrow \pi/2} \frac{(1 - \sin x) \sin^2 x}{(1 - \sin^2 x)}$$

$$\lim_{x \rightarrow \pi/2} \frac{(1 - \sin x) \sin^2 x}{(1 - \sin x)(1 + \sin x)}$$

$$\lim_{x \rightarrow \pi/2} \frac{\sin^2 x}{1 + \sin x}$$

$$\lim_{x \rightarrow \pi/2} \frac{1}{2}$$

$$\frac{1}{2} //$$

$$(vii) \lim_{x \rightarrow 0} \frac{2x \sin x}{\sec x - 1}$$

$$\lim_{x \rightarrow 0} \frac{2x \sin x}{(1 - \cos x)} \cos x$$

$$\lim_{x \rightarrow 0} \frac{x \sin 2x \times (1 + \cos x)}{(1 - \cos^2 x)}$$

$$\lim_{x \rightarrow 0} \frac{2x \sin x \cdot \cos x (1 + \cos x)}{\sin^2 x}$$

$$2 \lim_{x \rightarrow 0} \frac{x}{\sin x} \times \lim_{x \rightarrow 0} \cos x \times \lim_{x \rightarrow 0} (1 + \cos x)$$

$$2 \times 1 \times 1 \times 2$$

$$4 //$$

$$(ix) \lim_{x \rightarrow 0} \frac{\sin 3x + \sin x - 2 \sin 2x}{x \sin^2 x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin 2x \cos x - 2 \sin 2x}{x \sin^2 x}$$

$$2 \lim_{x \rightarrow 0} \frac{\sin 2x (\cos x - 1)}{x (1 - \cos^2 x)}$$

$$-4 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \times \lim_{x \rightarrow 0} \frac{(1 - \cos x)}{(1 - \cos x) \times (1 + \cos x)} = -4 \times 1 \times \frac{1}{2} = -2 //$$

Q. (i) ev.

$$\lim_{x \rightarrow \pi/4} \frac{\cos^5 x - \sin^5 x}{\cos 2x}$$

$$\lim_{x \rightarrow \pi/4} \frac{(\cos x - \sin x) (\cos^4 x + \cos^3 x \sin x + \cos^2 x \sin^2 x + \cos x \sin^3 x + \sin^4 x)}{(\cos x - \sin x) (\cos x + \sin x)}$$

$$\lim_{x \rightarrow \pi/4} \frac{5 \times \left(\frac{1}{\sqrt{2}}\right)^4}{\sqrt{2}} = \frac{5}{4\sqrt{2}} //$$

$$(x) \lim_{x \rightarrow \pi/4} \frac{\cos^5 x - \sin^5 x}{\cos 2x} \quad (0/0)$$

$$\lim_{x \rightarrow \pi/4} \frac{(5 \cos^4 x \cdot x - \sin x) - 5 \sin^4 x \cdot x \cos x}{- \sin 2x \times 2}$$

$$\lim_{x \rightarrow \pi/4} \frac{- \sin x \cdot 5 \cos^4 x - 5 \sin^4 x \cos x}{- 2 \sin 2x}$$

$$- 5 \lim_{x \rightarrow \pi/4} \frac{\sin x \cos^4 x + \sin^4 x \cos x}{- 2 \sin 2x}$$

$$= \frac{5}{2} \left[\lim_{x \rightarrow \pi/4} \frac{\sin x \cos^4 x}{\sin 2x} + \lim_{x \rightarrow \pi/4} \frac{\sin^4 x \cos x}{\sin 2x} \right]$$

$$= \frac{5}{2} \left[\lim_{x \rightarrow \pi/4} \frac{\sin x \cos^4 x}{2 \sin x \cos x} + \lim_{x \rightarrow \pi/4} \frac{\sin^4 x \cos x}{2 \sin x \cos x} \right]$$

$$= \frac{5}{2} \times \frac{1}{2} \left[\lim_{x \rightarrow \pi/4} \cos^3 x + \lim_{x \rightarrow \pi/4} \sin^3 x \right]$$

$$= \frac{5}{4} \left[\left(\cos \frac{\pi}{4} \right)^3 + \left(\sin \frac{\pi}{4} \right)^3 \right]$$

$$= \frac{5}{4} \left[\frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \right] = \frac{5}{4} \times \frac{1}{\sqrt{2}} \times 2 = \frac{5}{4\sqrt{2}}$$

$$(x') \lim_{x \rightarrow \pi/4} \frac{1 - \sin 2x}{\cos x - \sin x}$$

$$\lim_{x \rightarrow \pi/4} \frac{1 - \cos(\pi/2 - 2x)}{\sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x \right)}$$

$$\lim_{x \rightarrow \pi/4} \frac{2 \sin^2(\pi/4 - x)}{\sqrt{2} \sin(\pi/4 - x)}$$

$$\times \frac{1}{\sqrt{2}} \times 2 \times \lim_{x \rightarrow \pi/4} \sin(\pi/4 - x)$$

$$\sqrt{2} \times \sin(0)$$

0//

$$(x'') \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{\sin x - \sin x \cos x}{x^3 \times \cos x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{x^3 \cos x}$$

$$\lim_{x \rightarrow 0} \frac{1}{\cos x} \times \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \times \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$1 \times 1 \times \lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{x^2}$$

$$1 \times 1 \times 2 \left(\lim_{x \rightarrow 0} \frac{\sin x/2}{x/2} \right)^2$$

$$\frac{1}{2} //$$

$$(xiii) \lim_{x \rightarrow 0} \frac{1 - \cos ax}{1 - \cos bx}$$

$$\lim_{x \rightarrow 0} \frac{\cancel{x} \sin^2 ax/2}{\cancel{x} \sin^2 bx/2}$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 ax/2}{(ax/2)^2}$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 (bx/2)}{(bx/2)^2}$$

$$\lim_{x \rightarrow 0} \frac{a^2 \times 1 - a^2}{b^2} = \frac{a^2}{b^2} //$$

$$(xiv) \lim_{x \rightarrow 0} \frac{x}{\sqrt{1 - \cos x}}$$

$$\lim_{x \rightarrow 0} \frac{x}{\sqrt{2 \sin^2 x/2}}$$

$$\frac{1}{\sqrt{2}} \lim_{x \rightarrow 0} \frac{x}{\sin x/2}$$

$$\sqrt{2} \lim_{x \rightarrow 0} \frac{\sin x/2}{x/2}$$

$$\sqrt{2} //$$

$$(xv8) \quad \lim_{x \rightarrow \pi/3} \frac{1 - 2\cos x}{\sin(x - \pi/3)} \quad (\frac{0}{0}) \times$$

$$\lim_{x \rightarrow \pi/3} \frac{2 \sin x}{\cos(x - \pi/3)}$$

$$2 \lim_{x \rightarrow \pi/3} \sin x$$

$$\lim_{x \rightarrow \pi/3} \cos(x - \pi/3)$$

$$2 \times \frac{(\frac{\pi}{3})}{2} = \sqrt{3} //$$

$$(xv9) \quad \lim_{x \rightarrow 0} \frac{\sin(a+x) - \sin(a-x)}{x}$$

$$\lim_{x \rightarrow 0} \frac{2 \cos\left(\frac{a+x+a-x}{2}\right) \times \sin\left(\frac{a+x-a-x}{2}\right)}{x}$$

$$2 \lim_{x \rightarrow 0} \frac{\cos(a) \sin x}{x}$$

$$2 \cos a \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$2 \cos a //$$

$$(xvii) \quad \lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a}$$

$$\lim_{x \rightarrow 0} \frac{2 \cos\left(\frac{x+a}{2}\right) \sin\left(\frac{x-a}{2}\right)}{(x-a)}$$

$$\lim_{x \rightarrow 0} \cos\left(\frac{x+a}{2}\right) \lim_{x \rightarrow 0} \frac{\sin\left(\frac{x-a}{2}\right)}{\left(\frac{x-a}{2}\right)}$$

$$\cos \frac{a}{2} \times 1 = \cos \frac{a}{2}$$

$$(xviii) \quad \lim_{x \rightarrow 0} \frac{(a+x)^2 \sin(a+x) - a^2 \sin a}{x}$$

$$\lim_{x \rightarrow 0} \frac{(a^2 + x^2 + 2ax) \sin(a+x) - a^2 \sin a}{x}$$

$$\lim_{x \rightarrow 0} \frac{a^2 [\sin(a+x) - \sin a] + \lim_{x \rightarrow 0} \frac{\sin(a+x) x^2}{x} + 2a \lim_{x \rightarrow 0} \frac{\sin(a+x) x}{x}}{x}$$

$$\lim_{x \rightarrow 0} \frac{a^2 \cdot 2 \cos\left(\frac{2a+x}{2}\right) \sin\left(\frac{x}{2}\right) + \lim_{x \rightarrow 0} x \sin(a+x) + 2a \times \sin a}{x}$$

$$\frac{a^2 \lim_{x \rightarrow 0} \cos\left(\frac{2a+x}{2}\right) \times \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} + \lim_{x \rightarrow 0} x \sin(a+x) + 2a \sin a}{1}$$

$$\frac{a^2}{2} \times \cos(a) \times 1 + \lim_{x \rightarrow 0} x \sin(a+x) + 2a \sin a.$$

$$\frac{a^2}{2} (\cos a + 2a \sin a + \lim_{x \rightarrow 0} x \sin(a+x))$$

$$\frac{a^2}{2} (\cos a + 2a \sin a + \frac{0}{0} \times \sin a)$$

$$\frac{a^2}{2} \cdot \frac{a(\cos a + 2a \sin a)}{2} = \frac{a^3}{4} (\cos a + 2a \sin a)$$

$$(\sin) \text{ (ii) } \lim_{x \rightarrow \pi/4} \frac{\sin x - \cos x}{x - \pi/4}$$

$$\lim_{x \rightarrow \pi/4} \left(\frac{\frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x}{(x - \pi/4)} \right)$$

$$\lim_{x \rightarrow \pi/4} \left(\frac{\sin x \cos \pi/4 - \cos x \sin \pi/4}{(x - \pi/4)} \right)$$

$$\lim_{x \rightarrow \pi/4} \frac{\sin(x - \pi/4)}{(x - \pi/4)} = \lim_{x \rightarrow \pi/4} \frac{\sin u}{u} = 1$$

$$20) \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 - \sin x + \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{(1 + \sin x)^2 - \cos^2 x}{(1 - \sin x)^2 - \cos^2 x} \cdot \frac{(1 - \sin x + \cos x)}{(1 + \sin x + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{(1 + 2\sin x + \sin^2 x - \cos^2 x)(1 - \sin x + \cos x)}{(1 - 2\sin x + \sin^2 x - \cos^2 x)(1 + \sin x + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos^2 x + 2\sin x + \sin^2 x)(1 - \sin x + \cos x)}{(1 - \cos^2 x + \sin^2 x - 2\sin x)(1 + \sin x + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{(2\sin x + 2\sin^2 x)(1 - \sin x + \cos x)}{(2\sin^2 x - 2\sin x)(1 + \sin x + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{2\sin x (\sin x + 1)(1 - \sin x + \cos x)}{2\sin x (\sin x - 1)(1 + \sin x + \cos x)}$$

$$= \frac{(\sin 0 + 1)(1 - \sin 0 + \cos 0)}{(\sin 0 - 1)(1 + \sin 0 + \cos 0)}$$

$$= \frac{(0+1)(1-0+1)}{(0-1)(1+0+1)}$$

$$= -1 //$$

$$(22) \lim_{h \rightarrow 0} \frac{\sin(a+3h) - 3\sin(a+2h) + 3\sin(a+h) - \sin a}{h^3}$$

$$\left(\frac{0}{0}\right) \lim_{h \rightarrow 0} \frac{\cos(a+3h) \times 3 - 6\cos(a+2h) + 3\cos(a+h)}{3h^2}$$

$$\lim_{h \rightarrow 0} \frac{\cos(a+3h) - 2\cos(a+2h) + \cos(a+h)}{h^2}$$

$$\lim_{h \rightarrow 0} \frac{-3\sin(a+3h) + 4\sin(a+2h) - \sin(a+h)}{2h}$$

$$\lim_{h \rightarrow 0} \frac{-9\cos(a+3h) + 8\cos(a+2h) - \cos(a+h)}{2}$$

$$= \cos a //$$

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$$(23) \lim_{x \rightarrow 0} \frac{\sin 3x}{(\sqrt{x+2} - \sqrt{2})}$$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{(x+2 - 2)} \cdot (\sqrt{x+2} + \sqrt{2})$$

$$3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \lim_{x \rightarrow 0} (\sqrt{x+2} + \sqrt{2})$$

$$3 \times 1 \times 2\sqrt{2}$$

$$6\sqrt{2} //$$

$$x \rightarrow 0$$

$$\frac{1}{x}$$

14

(xxi) $\lim_{x \rightarrow 0} \frac{1 - \cos 3(\sin x)}{1 - \cos^2 x}$

sol

$$\frac{\lim_{x \rightarrow 0} 2 \sin^2 \left(\frac{\sin x}{2} \right)}{\sin^2 x}$$

$$\frac{2}{4} \lim_{x \rightarrow 0} \frac{\sin^2 \left(\frac{\sin x}{2} \right)}{\sin^2 x}$$

$$\frac{1}{2} \times \left[\lim_{x \rightarrow 0} \frac{\sin \left(\frac{\sin x}{2} \right)}{\left(\frac{\sin x}{2} \right)} \right]^2$$

$$\frac{1}{2} \times 1 = \frac{1}{2}$$